

Advanced Machine Learning and Deep Learning Approaches for Terrain Detection and Segmentation for Autonomous Vehicle Navigation

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Abstract

Although deep learning-based terrain segmentation models perform well on structured urban benchmarks, domain shift and dataset bias cause their dependability to drastically deteriorate when used in unstructured off-road conditions. This limitation poses a major challenge for autonomous navigation systems operating outside controlled settings. This study investigates whether lightweight domain adaptation and usage of multimodal techniques can improve the cross-domain generalization of terrain segmentation models. A pretrained DeepLabV3+ model with a ResNet-50 backbone, pre-trained on ImageNet is trained on the Cityscapes dataset and evaluated on an off-road RUGD terrain dataset, where a substantial performance drop is observed. Domain adaptation is then performed by partially fine-tuning the model on a limited subset of target-domain images, resulting in notable performance improvements. Additionally, a simple multimodal extension using edge-map inputs is explored to assess its impact on cross-domain robustness. Experimental results show that lightweight domain adaptation significantly reduces cross-domain performance degradation, however multimodal inputs provide more localized and inconsistent improvements. These results offer useful insights for enhancing terrain segmentation robustness in real-world autonomous navigation scenarios and show the advantages and disadvantages of resource-efficient adaptation strategies.

Keywords: Autonomous vehicles, Terrain detection, Segmentation techniques, Machine learning, Deep learning, Computer vision

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Chapter 1: Introduction

1.1 Background Study

In order to identify accessible surfaces and hazardous areas necessary for safe course planning and control, terrain segmentation is a fundamental perception function for autonomous vehicles [1]. When trained on large-scale benchmarks like Cityscapes and KITTI, deep learning-based techniques, in particular convolutional neural networks (CNNs) and transformer-based architectures, have demonstrated great performance in structured urban environments [2]. Consistent visual signals, clearly defined road features, and abundant annotations are advantageous to these models, which are frequently further improved by multimodal inputs like LiDAR for geometric reasoning [2][3]. However, performance significantly deteriorates when used in unstructured off-road conditions with a variety of terrain types, limited geometric signals at long ranges, and highly changing illumination [1]. The fundamental cause of this degradation is domain shift, which makes learned representations unreliable for safety-critical navigation tasks since the statistical characteristics of training data differ greatly from those found during deployment [4].

A major contributor to this domain shift is dataset bias in existing terrain segmentation benchmarks. The majority of research is heavily focused on urban-focused datasets, which implicitly embed assumptions about reliability and infrastructure regularity that are not true in natural settings [4][5][6]. While a number of off-road datasets offer useful multimodal annotations specific to unstructured terrains, their scope and variety are still restricted in comparison to urban datasets [5]. Real-world failures in commercial off-road autonomous systems, agriculture, and the military are caused by this mismatch, which limits model generalization [5]. Additionally, it is expensive and difficult to gather and annotate off-road data, necessitating specialist platforms and professional classification of confusing terrain classes [5][6]. The long tail distribution of rare but safety-critical features, such as concealed ditches or unstable slopes, further exacerbates the problem, as these events are insufficiently represented in limited datasets yet have a disproportionate impact on operational safety [7].

Domain adaptation techniques have emerged as a promising solution to mitigate urban to off road performance gaps, but their practicality is constrained by computational and data limitations inherent to autonomous vehicle platforms [8][9]. Conventional unsupervised domain adaptation methods often rely on adversarial training, large-scale offline optimization, or extensive target domain data, making them unsuitable for real-time or resource-constrained deployment [10]. By utilizing self-supervision and lightweight adaptation techniques, on the other hand, they have shown more viability. Methods such as real-time adaptation using near-range LiDAR supervision or prototype-based online updates enable continuous learning with minimal memory and computational overhead [11]. These tactics are especially pertinent for off-road vehicles running on embedded hardware with power, memory, and latency constraints, where adaptation must take place within constrained real-time budgets without compromising navigation safety [10][11].

Beyond domain adaptation, multimodal sensor fusion provides another effective way to improve robustness when facing domain shifts and limited training data. By combining different sensing modalities, systems can leverage their complementary strengths: cameras capture rich semantic

information but are highly sensitive to changes in appearance, while LiDAR delivers accurate geometric measurements at close range, though its data becomes sparse at longer distances and lacks semantic detail [12]. By combining these modalities, fusion-based systems reduce dependence on any single sensor’s domain-specific biases. Prior work has shown that integrating camera and LiDAR data in unified representations can significantly improve traversability estimation and semantic understanding in challenging off-road environments such as deserts, forests, and rocky terrain [12][13]. Multimodal approaches have also shown resilience in adverse conditions such as snow or low visibility, where individual sensors degrade unevenly.

Despite their advantages, existing multimodal fusion methods often rely on complex architectures that increase computational cost and hinder deployment on edge platforms. Large transformer encoders, graph-based models, and attention methods increase representational power but are incompatible with the efficiency demands of autonomous systems in the actual world [14][15][16]. Therefore, creating straightforward and lightweight fusion algorithms that successfully improve cross-domain generalization while adhering to stringent computational restrictions continues to be a crucial research need [15]. In order to close this gap and enable reliable terrain segmentation in unstructured contexts despite scarce data, substantial domain shift, and real-time operational demands, effective multimodal fusion must be carefully integrated with lightweight domain adaptation algorithms.

1.2 Problem Motivation

Recent advances in deep learning have enabled highly accurate terrain and semantic segmentation models, particularly within structured urban environments where road boundaries, object classes, and visual appearances are well-defined. Benchmark datasets such as Cityscapes have driven rapid progress in this domain, leading to models that perform reliably under controlled conditions. However, when these models are deployed in unstructured off-road environments, their performance often degrades substantially due to severe domain shift, appearance variability, and dataset bias. Off-road terrains exhibit irregular surfaces, ambiguous boundaries, and rapidly changing visual characteristics that are not adequately represented in urban training data, limiting the generalization capability of existing models.

This performance gap presents a critical challenge for autonomous systems operating in real-world off-road scenarios, where reliable terrain perception is essential for safe navigation. In practice, mitigating this gap often requires environment-specific retraining, extensive manual annotation, or continuous human supervision, all of which undermine the autonomy and scalability of such systems. Moreover, off-road autonomous platforms are frequently constrained by limited onboard computation and energy budgets, necessitating real-time processing on embedded hardware [17][18]. Consequently, many state-of-the-art solutions that rely on large models, extensive labeled datasets, or computationally intensive training procedures are impractical for deployment.

These challenges motivate the exploration of lightweight and resource-efficient strategies that can improve cross-domain generalization without imposing substantial computational or data collection overheads. In particular, domain adaptation techniques that leverage small amounts of target-domain data,

as well as simple multimodal cues derived from low-cost preprocessing, present promising directions for enhancing terrain segmentation robustness under practical deployment constraints.

1.3 Importance and relevance

Addressing the challenges of terrain segmentation under domain shift is critical for enabling reliable autonomy in a wide range of real-world applications, including agriculture, search and rescue, environmental monitoring, and defense. Improved segmentation robustness directly translates to safer navigation, reduced system failures, and increased operational coverage in environments where human supervision is limited or unavailable. As autonomous systems are increasingly expected to function beyond well-structured urban settings, the ability to generalize across diverse terrains becomes a key enabler of broader adoption.

From a research perspective, this work is relevant to advancing the state of the art in efficient perception for autonomous systems. By focusing on lightweight adaptation and fusion strategies, the study aligns itself with the growing emphasis on deployable and resource-aware machine learning solutions. The outcomes of the study have the potential to inform future dataset design, model architectures, and adaptation frameworks. This is contributing to more robust and practical terrain segmentation systems capable of operating under real-world constraints.

1.4 Research objective

The primary objective of this study is to evaluate whether lightweight domain adaptation techniques and simple multimodal feature integration can improve the cross-domain generalization of deep learning-based terrain segmentation models when transferring from structured urban environments to unstructured off-road terrains.

Specifically, this research aims to:

1. To quantify the degree of performance degradation encountered when urban-trained segmentation models are directly applied to off-road environments.
2. To investigate the effectiveness of fine-tuning using a small subset of target-domain images as a lightweight domain adaptation strategy.
3. To examine the impact of incorporating an auxiliary, low-cost modality (such as edge maps) on segmentation robustness under domain shift.
4. To analyze the combined and individual contributions of domain adaptation and multimodal input under constrained data and computational settings.

By focusing on minimal interventions and realistic deployment constraints, this study seeks to provide practical insights into improving terrain perception robustness for autonomous navigation systems without relying on extensive retraining or additional sensing hardware.

1.5 Paper outline

The remainder of the paper is organized as follows. Section 2 provides a literature review of existing research on terrain segmentation, domain adaptation, and multimodal learning, highlighting their strengths and limitations. Section 3 describes the proposed methodology, including dataset preparation, model architecture, and the experimental workflow. Section 4 presents the experimentation, implementation details, and results, including quantitative metrics and qualitative analysis. Section 5 discusses the findings, explaining why the proposed approaches improve cross-domain generalization and comparing results with existing work. Finally, Section 6 concludes the paper, summarizing contributions, limitations, and future directions for enhancing terrain segmentation in autonomous navigation.

Chapter 2: Literature Review

2.1 Overview

Terrain segmentation plays a key role in autonomous navigation, as it directly supports path planning and obstacle avoidance. Substantial progress has been made over the past two decades, largely due to improvements in sensing hardware, the availability of benchmark datasets, and advances in deep learning-based perception. Most of this progress, however, has focused on structured urban environments, where road layouts, visual appearance, and semantic categories are relatively stable. In contrast, off-road terrain introduces significantly greater variability, along with pronounced domain shift and a scarcity of labeled training data.

This chapter reviews prior work on terrain segmentation and related perception tasks. The discussion covers traditional and learning-based terrain perception methods, deep learning models and datasets for autonomous driving, approaches to off-road terrain understanding and traversability estimation, multimodal sensor fusion, and challenges related to domain adaptation and generalization. By focusing on these areas, the study can identify limitations in existing approaches and consequently choose the methods explored in this study.

2.2 Early Approaches to Terrain Detection and Off-Road Perception

Early work on off-road terrain perception relied heavily on geometric reasoning and handcrafted features from classical computer vision. One of the earlier systems, proposed by Manduchi et al. (2005), combined stereo vision with laser range data to detect obstacles and classify terrain in outdoor environments [19]. These approaches used manually designed cues such as surface roughness, slope, color histograms, and geometric consistency to distinguish between traversable and non-traversable regions. While effective in relatively constrained settings, they depended on strong assumptions about terrain structure and sensor calibration, which limited their robustness in highly unstructured environments with vegetation, irregular surfaces, and varying illumination conditions [19].

As the field progressed, learning-based approaches gradually replaced rule-based pipelines with the adoption of probabilistic models and classical machine learning techniques. Surveys by Kabir et al., later summarized in SN Applied Sciences (2021), describe a shift toward data-driven terrain classification using features extracted from vision and range sensors [1][20]. These methods improved robustness compared to handcrafted systems but still relied on carefully designed feature representations and were often trained and evaluated within a single environment. Consequently, generalization across different terrains, seasons, and geographic regions remained limited.

Despite these improvements, early learning-based methods continued to struggle under distribution shift. Many representations were heavily engineered and tailored to specific environments, sensor configurations, or terrain types [21]. Models trained in one geographic region or season frequently performed poorly when deployed elsewhere. This lack of robustness under distribution shift directly

motivates the domain adaptation problem explored in this study, where models trained on structured urban data fail to generalize to off-road terrain.

2.3 Deep Learning–Based Segmentation for Autonomous Driving

The introduction of deep convolutional neural networks marked a major shift in semantic segmentation for autonomous driving. CNN-based architectures replaced hand-crafted features with learned hierarchical representations, allowing models to capture complex spatial and semantic patterns directly from data [22]. Early successes in image classification quickly extended to dense prediction tasks, leading to the development of fully convolutional networks and encoder–decoder architectures for pixel-level segmentation [23].

Among these approaches, the DeepLab introduced atrous convolution and atrous spatial pyramid pooling to capture multi-scale contextual information while preserving spatial resolution [24]. DeepLab variants achieved strong performance on benchmark datasets such as Cityscapes and PASCAL VOC, which established them as standard models for road scene segmentation [25]. Their ability to model both local detail and global context influenced the architecture choice in this work. Progress in this area was further supported by the availability of large-scale annotated datasets. Cityscapes provided high-quality pixel annotations for urban scenes, while KITTI and SemanticKITTI expanded benchmarks to include depth and LiDAR-based segmentation tasks [26][27]. Together, these datasets standardized evaluation and made performance improvements in urban perception systems.

Despite these advances, multiple studies have shown that models trained on these benchmarks experience substantial performance degradation when applied outside their training distribution [28]. Variations in texture, illumination, and scene geometry contribute to poor generalization, particularly in unstructured environments. This limitation directly motivates the cross-domain evaluation performed in this study.

2.4 Off-Road Terrain Segmentation and Traversability Estimation

Off-road environments present challenges that are fundamentally different from those found in urban driving scenarios. Terrain geometry is often irregular, semantic boundaries are less clearly defined, and visual appearance varies widely even within the same class [29]. These factors make dense semantic labeling significantly more difficult and reveal the limits of segmentation models trained exclusively on structured urban datasets.

To address this gap, datasets such as RELLIS-3D were introduced to provide pixel-level annotations for natural and rural terrain [30][31]. Evaluation efforts like OFFSEG showed that models trained on Cityscapes suffer severe performance drops when tested on off-road data, underscoring the combined effects of domain shift and class imbalance [32]. In parallel, a separate line of work has focused on traversability estimation rather than semantic segmentation. These methods typically predict terrain cost or navigability using geometric cues or proprioceptive feedback [33]. While effective for motion planning, they often trade away semantic detail and are frequently designed for specific vehicle platforms.

More recent studies have explored transfer learning for traversability prediction, showing that pretrained models can reduce labeling effort in new environments [34]. Many of these approaches, however, depend on specialized sensors or require careful vehicle-specific calibration. This work instead targets RGB-based semantic segmentation under limited supervision, aiming for a solution that is easier to deploy and more broadly applicable across off-road settings.

2.5 Multimodal Sensor Fusion for Terrain Understanding

Multimodal perception has been widely explored as a way to improve robustness in difficult environments by combining complementary sensor modalities [35]. Camera images provide rich semantic cues, while LiDAR offers accurate geometric structure, which makes their combination effective for many perception tasks. Prior work has proposed a range of fusion strategies, including early fusion at the input level, mid-level feature fusion, and late fusion at the decision stage [36]. More recently, transformer-based fusion models have introduced cross-modal attention mechanisms that adaptively integrate information across modalities and report strong performance in structured driving scenarios [37].

In practice, fusion-based systems also introduce additional sources of fragility. Their performance depends heavily on accurate sensor calibration and synchronization, and they can degrade significantly when one modality becomes unreliable [38]. Moreover, most fusion approaches are evaluated primarily in urban settings and rely on expensive sensors such as LiDAR or radar, which limits their applicability to lightweight or resource-constrained platforms [39]. For these reasons, this study focuses on RGB-based auxiliary cues, such as edge maps and grayscale representations, as a lower-cost alternative that can still provide useful complementary information.

2.6 Lightweight and Weakly Supervised Alternatives

Practical deployment constraints have motivated growing interest in lightweight perception models that balance accuracy with computational efficiency [40]. Many autonomous platforms operate under tight power and processing limits, which makes large-scale architectures difficult to deploy in practice. To reduce annotation effort, weakly supervised semantic segmentation methods have been proposed that rely on image-level labels or partial supervision [41]. While these approaches can work in constrained settings, they often struggle with boundary accuracy and incomplete coverage, particularly for amorphous terrain classes. LiDAR-guided weak supervision has been used to improve segmentation quality, but most existing work targets object-centric tasks and depends on additional sensing hardware [42].

Parameter-efficient fine-tuning strategies offer a more practical alternative. By freezing pretrained backbones and adapting only task-specific layers, models can achieve meaningful performance improvements with limited data and modest computational cost [43]. This tradeoff between efficiency and adaptability aligns directly with the approach adopted in this study.

2.7 Domain Shift, Generalization, and Adaptation

Domain shift remains a central challenge in autonomous perception, as models often fail to generalize when deployed outside their training distribution [44]. This problem has been widely studied in settings such as adverse weather, nighttime driving, and cross-city transfer. Domain adaptation methods attempt to address these issues through techniques such as feature alignment, style transfer, and adversarial learning [45][46]. Although effective, many of these approaches require extensive retraining or large amounts of unlabeled data, which limits their practicality in resource-constrained scenarios.

More recent work has emphasized parameter-efficient adaptation strategies that modify only small parts of a network [47]. Freezing pretrained backbones while adapting classifier layers has emerged as a practical solution when labeled target data is scarce. Other efforts have explored uncertainty-aware segmentation to identify unreliable predictions under domain shift [48]. While promising, these methods introduce additional complexity and are not yet commonly used in off-road terrain perception pipelines. This gap reinforces the motivation for the simpler adaptation strategies explored in this work.

2.8 Summary and Research Gap

Prior work points to several consistent patterns. Deep learning-based segmentation performs well in structured urban settings, but accuracy drops sharply in off-road environments where visual appearance and terrain structure are less predictable. Multimodal fusion has been shown to improve robustness, yet many approaches depend on expensive sensors and computationally heavy models. Across these settings, domain shift remains a primary obstacle to reliable real-world deployment.

Despite the breadth of existing research, relatively little attention has been given to systematically evaluating lightweight and resource-efficient approaches for cross-domain generalization from urban to off-road terrain. In particular, there is limited understanding of how far simple domain adaptation techniques and low-cost multimodal cues can go when only minimal labeled target-domain data is available and architectural complexity is kept low.

This study aims to fill this gap by experimentally examining lightweight domain adaptation and simple multimodal input integration within a controlled setting relevant to autonomous navigation systems.

Chapter 3: Proposed Methodology and System Design

3.1 Overview

This chapter outlines the methodology used to study cross-domain generalization in terrain segmentation when models are transferred from structured urban environments to unstructured off-road scenes. The design of the approach is guided by practical deployment constraints, including limited labeled data, restricted computational resources, and very minimal dependence on sensing hardware.

The methodology follows a staged experimental pipeline. First, a baseline semantic segmentation model is trained on a structured urban dataset and evaluated directly on an off-road dataset to quantify performance degradation caused by domain shift. Next, a lightweight domain adaptation strategy is applied through partial fine-tuning on a small subset of off-road images. Finally, a simple multimodal technique is introduced, such as an edge map, which acts as an auxiliary input modality. Each stage is evaluated independently to assess its contribution to improving cross-domain segmentation performance.

3.2 Model Architecture

DeepLabV3+ with a ResNet-50 backbone is selected as the base segmentation model due to its strong performance on urban benchmarks and widespread adoption in semantic segmentation research. The model is initialized using ImageNet-pretrained weights to leverage general-purpose visual features learned from large-scale data. To accommodate the reduced number of terrain classes considered in this study, the final classification layer is modified to output predictions for six semantic classes: 0: background, 1: drivable terrain, 2: vegetation, 3: obstacles, 4: sky, and 5: water. The remaining architecture remains unchanged to preserve pretrained representations and minimize computational overhead.

3.3 Datasets and Label Mapping

3.3.1 Urban Dataset

Cityscapes is used as the structured urban source domain. The dataset contains high-resolution street-view images with fine-grained semantic annotations. For this study, only classes relevant to terrain understanding, such as road, vegetation, obstacles, and sky, are retained. These classes are remapped to a reduced label set to ensure compatibility with the off-road dataset.

3.3.2 Off-road Dataset

The RUGD dataset serves as the unstructured target domain and contains images captured across a wide range of off-road environments. These scenes include diverse terrain types such as dirt, vegetation, obstacles, water, and sky. The original color-coded annotations are converted into class labels using a custom color-to-label mapping. Pixels that do not correspond to the selected terrain categories are excluded from training and evaluation.

All images and corresponding labels from both datasets are resized to a fixed spatial resolution to maintain consistency during training and evaluation.

3.4 Baseline Cross-Domain Evaluation

The baseline experiment evaluates the extent of domain shift between urban and off-road environments. In this setup, the DeepLabV3+ model is trained exclusively on Cityscapes and then evaluated directly on RUGD without any form of adaptation. This experiment establishes a lower-bound performance reference and highlights the limitations of direct model transfer.

Training is performed for a limited number of epochs using standard cross-entropy loss, with ignored labels excluded from gradient computation. No off-road images are used during training at this stage.

3.5 Lightweight Domain Adaptation

To reduce the impact of domain shift while maintaining efficiency, a lightweight domain adaptation strategy is employed. A small subset of labeled off-road images is selected to simulate a few-shot adaptation scenario. During adaptation, the backbone network is frozen and only the segmentation head is fine-tuned on the target-domain data.

Class-weighted cross-entropy loss is used to address severe class imbalance in the few-shot dataset. By limiting the number of trainable parameters, this approach reduces the risk of overfitting while allowing the model to adjust its decision boundaries to the target domain.

3.6 Multimodal Feature Integration

To further improve robustness under domain shift, a simple multimodal extension is introduced using edge-based structural cues. Edge maps are extracted from RGB images using Sobel filters and concatenated with the original RGB input to form a four-channel tensor.

The first convolutional layer of the network is modified to accept the additional input channel, and pretrained weights are partially reused to initialize the new layer. The backbone remains frozen during training, and only the segmentation head is updated in order to preserve previously learned representations.

3.7 Evaluation Metrics

Segmentation performance is evaluated using mean Intersection-over-Union (mIoU), computed across all non-ignored classes. This metric provides a balanced measure of performance under class imbalance and is widely used in semantic segmentation literature.

Chapter 4: Experimentation, Implementation & Results

4.1 Experimental Design and Objectives

The experiments are designed to systematically analyze the effects of domain shift in terrain segmentation and to evaluate lightweight strategies for improving performance under constrained resources. The primary goal is to assess how a model trained on structured urban scenes behaves when applied to unstructured off-road environments and whether limited target-domain supervision can mitigate this degradation.

A sequence of experiments is conducted, beginning with a baseline evaluation without adaptation, followed by few-shot domain adaptation, multimodal input experiments, and a combined approach. In addition, a semi-supervised self-training experiment is included to examine its feasibility under noisy off-road conditions. This staged design allows for direct comparison between methods while isolating the contribution of each component.

4.2 Datasets and Preprocessing

Cityscapes is used as the source domain and consists of high-resolution RGB images captured in urban environments under relatively consistent lighting and structural conditions. In contrast, RUGD represents off-road scenes characterized by irregular terrain geometry, natural obstacles, varying illumination, and significant texture diversity.

To ensure compatibility between datasets, class labels are remapped into six unified terrain categories: background, drivable terrain, vegetation, obstacles, sky, and water. This simplified taxonomy enables focused evaluation while reducing inconsistencies between annotation schemes.

All images are resized to a fixed resolution prior to training. Standard ImageNet normalization is applied to align with the pretrained backbone initialization.

4.3 Model Architecture and Training Configuration

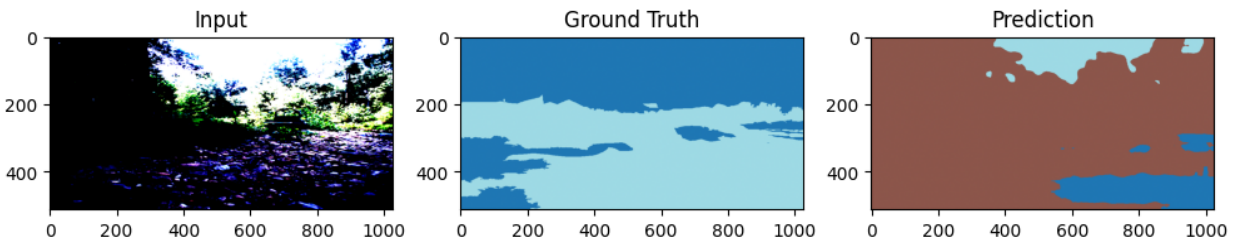
The DeepLabV3+ architecture with a ResNet-50 backbone is used throughout all experiments. Atrous spatial pyramid pooling allows the model to capture contextual information at multiple scales. The backbone is initialized using ImageNet-pretrained weights to provide robust low-level feature representations.

During baseline training, the entire network is trained on Cityscapes. For domain adaptation experiments, the backbone is frozen and only the classifier head is fine-tuned using RUGD samples. Cross-entropy loss is used for all experiments, with class-weighted loss applied when target-domain data is involved to address class imbalance, particularly for minority classes such as water and obstacles.

4.4 Baseline Cross-Domain Evaluation

In the baseline evaluation, the model trained on Cityscapes is tested directly on the RUGD dataset without adaptation. This experiment highlights the severity of the domain gap between urban and off-road environments. The model achieves a mean Intersection-over-Union of 0.0017, indicating near-complete failure to generalize.

Qualitative results show that predictions are dominated by incorrect or missing labels. The model relies heavily on urban-specific visual cues, such as road markings and building edges, which are largely absent in off-road scenes. The image below displays the ground truth and prediction to highlight the difference.

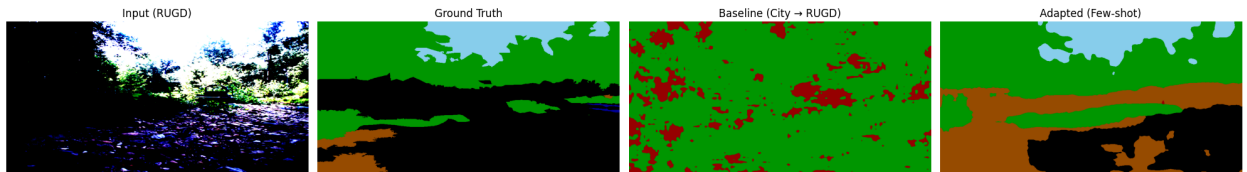


[Figure 1: Qualitative segmentation results of baseline model on RUGD]

4.5 Domain Adaptation

Few-shot domain adaptation is performed using small labeled subsets of the RUGD dataset. Experiments are conducted with 100, 150, and 200 training images to evaluate sensitivity to data availability. By freezing the backbone and fine-tuning only the classifier head, the model adapts to the target domain without disrupting previously learned representations.

This approach leads to a substantial improvement in performance, achieving a mean Intersection-over-Union of 0.2677. Improvements are most pronounced for drivable terrain and vegetation, which are critical classes for autonomous navigation.

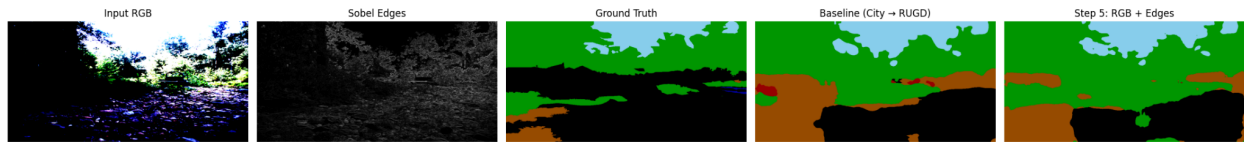


[Figure 2: Qualitative comparison before and after domain adaptation]

4.6 Multimodal Input Integration

To further address domain shift, additional low-level modalities are introduced in the form of edge maps and grayscale images. These representations are generated using standard image processing techniques and concatenated with the RGB input.

The motivation behind this approach is that structural cues, such as edges, are less sensitive to changes in color distribution and texture. Grayscale images similarly reduce reliance on color while emphasizing spatial structure. When trained using multimodal inputs without domain adaptation, the model achieves a mean Intersection-over-Union of 0.2630. While competitive, this approach does not consistently outperform domain adaptation across all classes.



[Figure 3 here: Edge maps and grayscale examples from RUGD images]

4.7 Combined Domain Adaptation and Multimodal Learning

In the final supervised experiment, few-shot domain adaptation and multimodal inputs are combined. The model is fine-tuned on the RUGD subset using RGB, edge, and grayscale channels as input. This configuration achieves the highest performance of all experiments, with a mean Intersection-over-Union of 0.3020. The results suggest that target-domain supervision and modality augmentation provide complementary benefits, allowing the model to better capture both semantic and structural information.

4.8 Self Training Using Pseudo Labels

A self-training strategy based on pseudo-labeling is explored as a semi-supervised alternative. The adapted model is used to generate predictions for unlabeled RUGD images, which are then treated as ground truth for additional training.

Although this approach improves over the baseline, the final mean Intersection-over-Union of 0.1525 remains well below that of supervised adaptation methods. Errors introduced in early pseudo-labels tend to propagate, particularly in visually ambiguous regions.

4.9 Consolidated Results

Consolidated Results	mIoU
Baseline	0.0017
Domain Adaptation	0.2677
Multimodal Only	0.2630
Combined domain and multimodal	0.3020

[Table 1: Consolidated Results]

Table 1 summarizes the performance of the different strategies under severe domain shift. The baseline model achieves an mIoU of only 0.0017, showing that models trained on structured urban data fail almost completely when applied to off-road terrain. Applying domain adaptation alone leads to a large improvement (mIoU = 0.2677), demonstrating that even limited labeled target data can significantly reduce the impact of domain shift.

Using multimodal inputs also improves performance (mIoU = 0.2630) by providing additional structural information, but this alone is not sufficient to fully bridge the domain gap. The best results are obtained when domain adaptation and multimodal inputs are combined, reaching an mIoU of 0.3020. Overall, these results show that effective off-road terrain segmentation benefits from both the leveraging adaptation strategies and auxiliary sensory information.

Chapter 5: Discussion

The results make it clear that domain shift is a major obstacle for terrain segmentation. Models trained on structured urban datasets perform almost no better than chance when applied to off-road environments, highlighting how poorly these representations transfer. This drop in performance is driven by both appearance differences and the inherent complexity of natural terrain, including irregular textures, changing illumination, and frequent occlusions.

Few-shot domain adaptation proved to be an effective way to address this gap. Even with very limited labeled data from the target domain, fine-tuning only the classifier head led to substantial performance gains. Keeping the backbone frozen helped prevent overfitting and preserved useful features learned from the source domain, suggesting that a small amount of target supervision can meaningfully recalibrate predictions without full retraining.

Adding multimodal inputs introduced complementary information, such as depth or infrared cues, that is not captured by RGB images alone. While multimodal inputs by themselves were not sufficient to overcome domain shift, combining them with domain adaptation produced the strongest results. This indicates that auxiliary modalities can improve feature alignment across domains, particularly when paired with even minimal labeled target data.

Self-training via pseudo-labeling was less successful in this setting. The high visual variability of off-road environments increases the risk of incorrect pseudo-labels, and without strong confidence filtering or refinement, these errors accumulate and hurt performance. This suggests that semi-supervised approaches must be applied carefully in challenging terrain scenarios.

Overall, the results show that handling domain shift in terrain segmentation requires a combination of targeted adaptation and additional sensory information, with close attention to the robustness of each component.

Chapter 6: Conclusion and Future Directions

6.1 Conclusion

This work examined terrain segmentation in the presence of severe domain shift, with a particular focus on transferring models from urban to off-road environments. The results show that direct transfer performs poorly, largely because of substantial differences in appearance, scene structure, and class distributions between the two domains. In contrast, lightweight adaptation strategies lead to meaningful improvements. Few-shot fine-tuning of the classifier head, while keeping the backbone frozen, was sufficient to adapt the model using only a small amount of target-domain data.

Performance was further improved by incorporating auxiliary modalities, which provide structural cues that are difficult to infer from RGB imagery alone, especially when combined with domain adaptation. Addressing class imbalance during training also played an important role, helping the model better recognize less frequent terrain classes. Together, these findings suggest that effective terrain segmentation for autonomous navigation does not require large-scale annotation but instead benefits from targeted, resource-efficient adaptation techniques.

6.2 Future Directions

There are several directions in which this work could be extended. Collecting more diverse labeled data from the target domain would likely improve adaptation and help the model generalize across a wider range of off-road conditions. More advanced domain adaptation methods, including adversarial feature alignment, style transfer, and contrastive learning, could further reduce the domain gap by encouraging shared representations between urban and off-road data.

Future work could also revisit self-training with stronger safeguards, such as confidence-based filtering, label refinement, or iterative pseudo-label updates, to limit the impact of noisy predictions. Incorporating geometric modalities, such as LiDAR data from datasets like RUGD, may provide valuable depth and structural information that complements visual inputs. Finally, evaluating these approaches in real-time autonomous navigation settings and expanding the set of terrain classes would help assess their practicality and robustness in real-world off-road applications.

Chapter 7: References

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